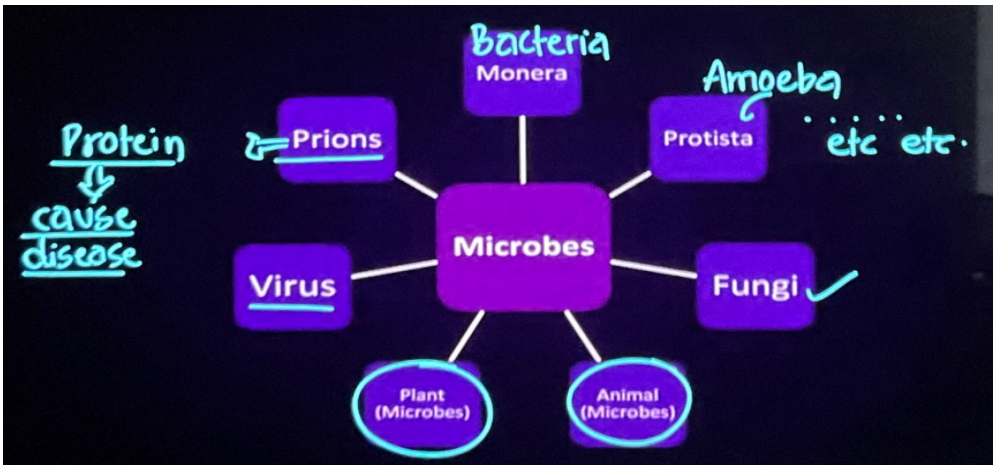
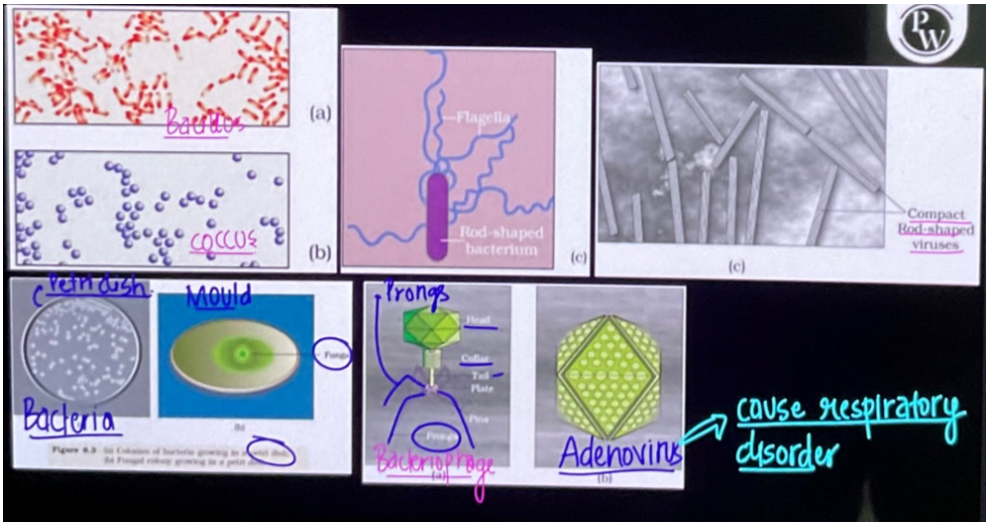
12th HACKERS KA JUGAAD

Microbes in Human Welfare



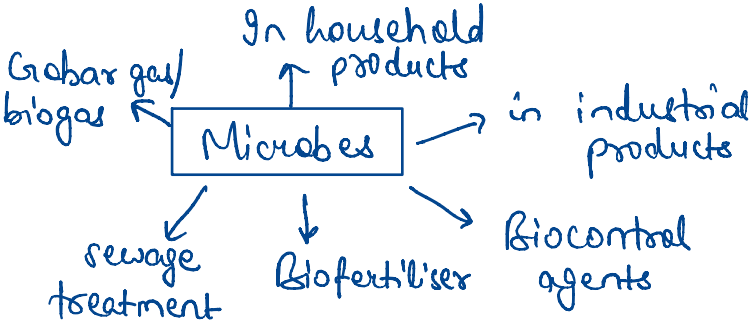


Microbes
↓
everywhere
- soil
- Air
- Water
- inside other organism
- ↓ where other spp. (life form not
- can be grown in synthetic medium, colonies

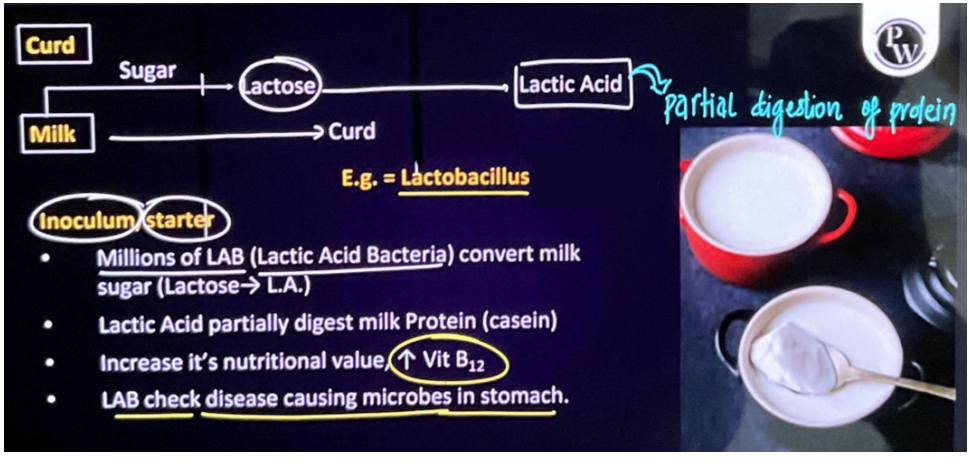
found)

can be seen naked eyes.

deep oceans | thermal vents | deep in soil | snow/ ice



Microbes in household products — curd
— Bread
— Idli / Dosa
— Toddy
— Cheese



Bread

- Agent = Fungi → Yeast → Saccharomyces cerevisiae (Baker's Yeast)
- Yeast = Alcoholic fermentation, CO_2 is released.
- Dough ← Yeast
- Sugar → Ethanol + CO_2 ⇒ Evaporate from dough
  ⇒ Makes it soft. And porous, c/a leavening.



Idli and Dosa

- Agent = Bacteria (Alcoholic fermentation)
- E.g. = Leuconostoc, Streptococcus

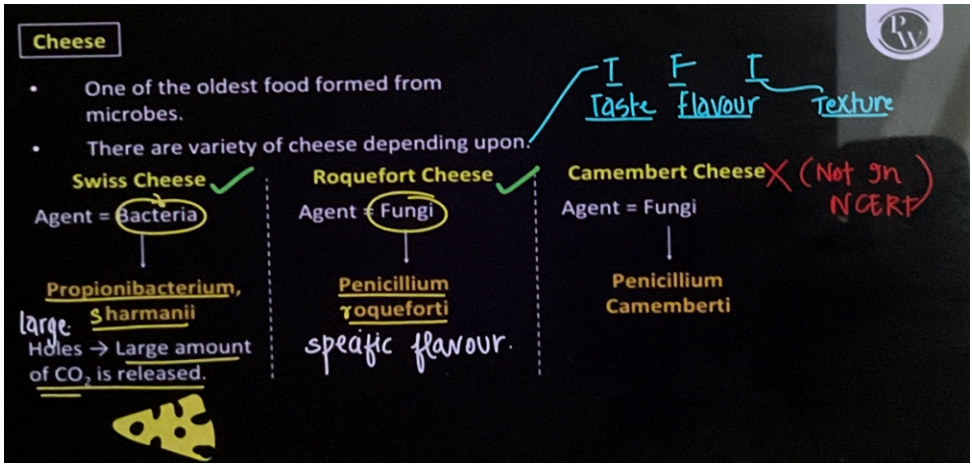
Rice / Urad Daal (Black Daal)
↓ Soaked Overnight
Grind
Batter ↑ Fluffy / Soft



Toddy

- Agent = Fungi (yeast – Alcoholic fermentation) naturally present.
- Collected from sap of palm tree. Caryota urens
- Traditional drink in many southern parts of India.





Microbes in Industrial products — fermented beverages

- antibiotics
- bioactive molecules
- organic acids
- enzymes.

Fermentors — large vessels, production at large scale. Capacity ~ 1000 litre.



Fermentation plant

fermented beverages – alcoholic drinks

Fermented Beverages (Alcoholic Drinks)

- Agent = Yeast = (Sacchoromyces cerevisiae) ⇒ Brewer's yeast
- Malted cereals *or raw* materials —(Yeast Fermentation)→ Fermented. beverages.
- Yeast poison itself to death when alcoholic content reaches about 13%.

Types depends on
- Processing (Distillation, Without distillation)
- Raw material

Distillation
- more alcoholic content
- >13%
- Whisky
  Brandy
  Gin
  Rum
  Vodka

Without distillation.
- lesser
- <13% (upto 13%)
- wine
- beer

Antibiotic –

Antibiotic — serendipity discovery.

Against Life ⇒ Bacteria (Pathogen)
Pro – life ⇒ Human

- First antibiotic was discovered by Alexander Flemming. (By mistake (chance serendipity))



Chemical was named = **Penicillin**
- First Discovered antibiotic.
- Full potential as antibiotic was discovered by Chain and Florey.

- Flemming
- Florey
- Chain

} Noble Prize in 1945.

- Penicillin was extensively used in World War-II.
- After penicillin several other antibiotics wee isolated from different agents. E.g. Streptomycin, Erythromycin, Chloramphinicol
- Treatment of many disease was possible :
  - Whooping cough (Kaali Khansi)
  - Diptheria (gal ghotu)
  - Leprosy (Kusht rog)

Microbe	Organic Acids
Acetobacter aceti (Bacteria)	Acetic Acid
Aspergillus niger (Fungi-Ascomycetes)	Citric Acid
Clostridium butylicum (bacteria)	Butyric Acid
Lactobacillus (Bacteria)	Lactic Acid

Enzymes -

lipase - digestion of oil and fat.
- added in detergents to remove oil stains

protease and pectinase - to clarify fruit juices

streptokinase - clot buster
- obtained from streptococcus
- given to patients = Myocardial infarction

Amylase - digestion of starch.



Bioactive Molecules

Cyclosporin-A

- agent = fungi
- trichoderma polysporum
- immunosuppressive agent given to organ transplant patients.

statin

- agent = fungi
- Monascus purpureus
- cholesterol lowering agent
- blood cholesterol ↓↓

Microbes as Biocontrol Agents

Biopesticide — Predators.

- Organisms are used to kill pest.
- Eradicate pest at manageable level, not completely.
- Helps integrated pest management
- Preferred in organic farming (sustainable)
- Ecologically sensitive (Diversity ↓↓, endangered) areas are treated with biopesticide

E.g.

(1) Butterfly Catterpilar on brassica plants other fruit plants.

Bacillus thuriengensis

- Release toxin → active inside gut of certain insects and kills it. (targeted)
- Sackets of dried spores of Bt

↓ mixed with H_2O

Sprinkle of leaves.

(2) Lady bird beetle — eats → Aphids

(3) Dragonflies — eats → Mosquito

(4) Baculovirus ⇒ genus = Nucleopolyhedrovirus ⇒ narrow spectrum specific insects and other arthropods, not effect on plants, animal, bird, mammals, non-targeted insects. (targeted insects)

Majorly

Trichoderma = free living fungus.



↑ Fertility of Soil. **Biofertiliser**

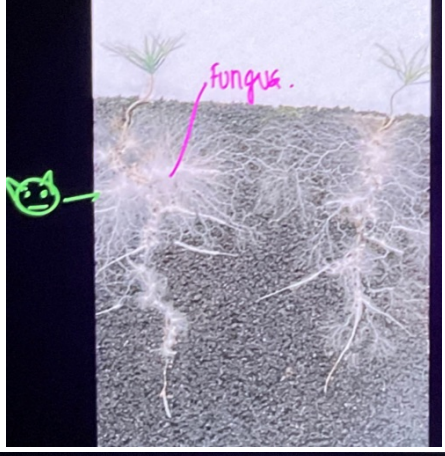
- Fungi
- Bacteria
- Cyanobacteria.

Fungi — **Mycorrhizal Association**

Roots of higher plant + Fungi Glomus

Pinus

- ↑ Surface area.
- Protect root from soil borne-pathogen
- Absorption of H_2O and minerals
- Make plant drought resistant.
- Absorption of phosphorus
- Make plant salt resistant.



N_2 – Fixing Cyanobacteria = enrich nutrient quality of salt.

E.g. Anabaena, Nostoc, Oscillatoria

- ↑ N-content in soil.
- Important biofertilizer in paddy field.
- Available in markets to replenish soil quality, to reduce dependency on chemical fertilisers.

N_2 – Fixing Bacteria

Symbiotic: Rhizobium

Free-Living: Azospirillum, Azoto bacter

↑ Nitrogen content in soil.

Sewage treatment plants

Sewage:

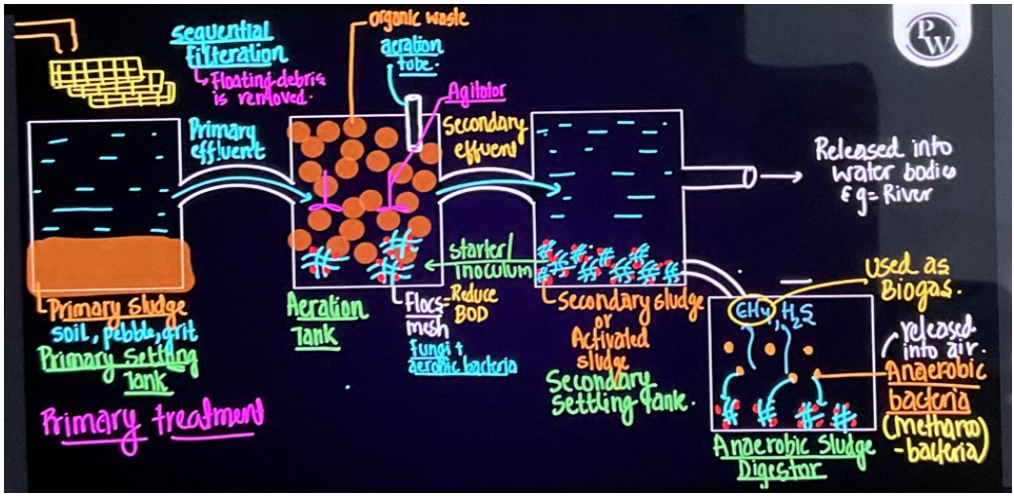
- municipal waste
- daily life was from house, schools, coachings (Human excreta)
- majorly organic waste
- Microbes (heterotrophic)
  - (some) Pathogens → diseases causing
  - (some) Digestion of molecules

Treatment

- Primary treatment
- Secondary treatment

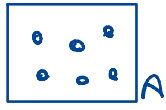
or
physical treatment

or
biological treatment

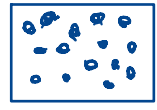


BOD = biochemical oxygen demand.

- oxygen required by bacteria in 1 litre of water to remove organic waste.



BOD↓



BOD↑

BOD↑↑ Pollution↑↑

BOD↓↓ Pollution↓↓

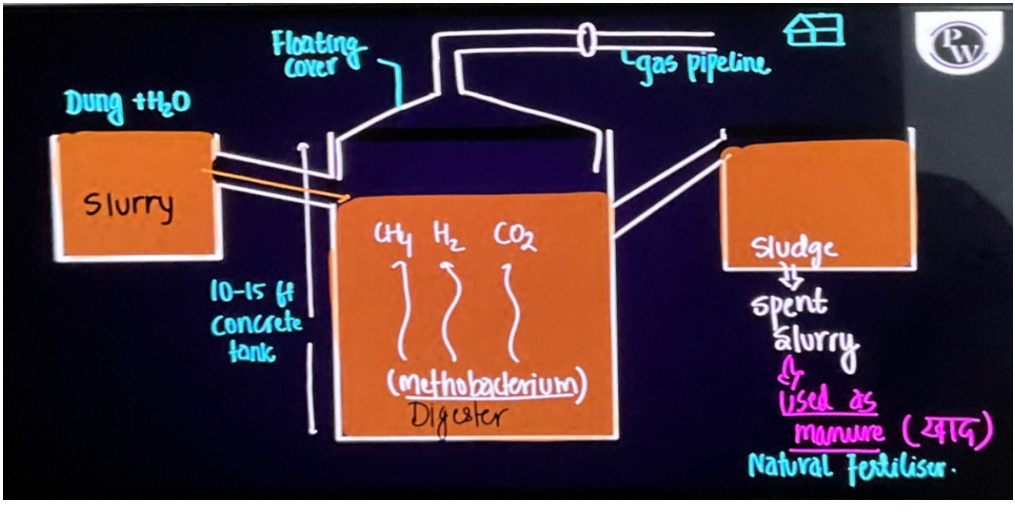
India ⇒ ministry of environment and forest.

Ganga Action Plan — Yamuna action Plan

Goal:
- to prevent rivers from water pollution
- to increase sewage treatment plants.

Microbes in biogas production

- Mixture of gas (predominantly CH_4) used as fuel.
- Microbe = Archaebacteria. (anaerobic)
- Methanogen (E.g. = Methanobacteria)
  - ↓
  - Found in anaerobic sludge digester in STP.
  - Rumen of cattles. part of stomach.
  - Obligate anaerobic.
- Present in Gobar, digestion of cellulose, release of CH_4 which is used as fuel c/a Gobar gas/Biogas.



In India,

IARI – Indian agricultural research institute

KVIC – Khadi and village industries commission

Biogas Plant mainly rural areas.

